🐄 Dairy: Guided Notes · ANSWER KEY

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

Use the class slides and your reading. Key words:

Holstein | *Jersey* | *colostrum* | *oxytocin* | *milking parlor* | *pasteurize* | *homogenize* | *casein* | *overrun*

A. Dairy production (Blank Out notes)

1. Milk is Washington's # **2** farm product.
2. The black-and-white breed with the most milk is the **Holstein**.
3. Milk let-down is caused by the hormone **oxytocin**.
4. **Percent practice**

> Example: milk is 87% water, so 10 lb of milk has 10 × 0.87 = 8.7 lb of water

a) Water in 100 lb of milk = **87 lb**

b) A herd of 100 cows gives 2,300,000 lb a year. Pounds per cow = **23,000 lb**

B. Farm to plant

5. Fill in the table.

Step	What happens
Truck check	**under 40°F, smell, sample tested**
Pasteurize	**heat (161°F, 15 s) kills harmful bacteria**
Homogenize	**breaks up fat so cream doesn't rise**

6. Why must pasteurized milk stay in the fridge?
Pasteurizing does not sterilize; some bacteria remain and grow fast when warm.

C. Dairy products (Cheese, Greek yogurt, Ice cream)

7. Cheese forms when **casein** clumps together.
8. Ice cream must have at least **10** % milk fat.
9. How is Greek yogurt different from regular yogurt?
It is strained to remove liquid, so it is thicker (and higher in protein).

🔑 *Key written by our team from Mr. Torretta's materials, please check.*